

Functional Equations - Strategic Substitution

State the domain. Choose an input for structural payoff, pair or combine equations when the argument map closes, prove the rule for all allowed inputs, then check the original equation.

Functional Equations: Choose Inputs with a Purpose

A functional equation tells you something about the same function at several related inputs. The hard part is usually not algebra after the equations are written. The hard part is choosing the input that makes the structure collapse.

The central habit is:

> Do not ask "which number should I try?" Ask "which input removes the most structure?"

1. Start with the legal input set

Before substituting anything, read the domain.

If the equation is stated for all integers, then 0, 1, -1, and integer expressions are legal. If it is stated for all real numbers, a partner such as $3-x$ is legal for every real x .

A substitution is invalid if it leaves the allowed input set.

2. Special values: make a complicated argument simple

Suppose an equation contains $mn+1$. One variable equal to 0 turns that argument into 1.

That is more useful than trying $m=2$ merely because 2 is small.

Example

For integers m, n , suppose

$$F(m+n)=F(m)+n$$

and $F(0)=4$.

Set $m=0$:

$$F(n)=F(0)+n=n+4.$$

One substitution determines the whole function on the stated integer domain.

The point is not that zero is magical. It is that zero collapses the structure.

3. Return-partner substitution

A transformation is especially useful when applying it twice returns to the original input.

For example,

$$x \rightarrow 5-x \rightarrow 5-(5-x)=x.$$

If an equation contains both $f(x)$ and $f(5-x)$, write it once at x and once at $5-x$. This kind of self-reversing partner map is sometimes called an involution; the useful idea is the return, not the vocabulary.

Example

For every real x ,

$$2f(x)+f(3-x)=x+9.$$

Write the partner equation by replacing x with $3-x$:

$$2f(3-x)+f(x)=12-x.$$

Now the two unknowns are just $f(x)$ and $f(3-x)$. Solve the 2×2 system:

$$f(x)=x+2.$$

The function was not guessed. It was forced by the pair.

4. Combine equations to eliminate a companion value

After you create a partner equation, the next move is ordinary algebra: add, subtract, or eliminate.

The algebra itself belongs to familiar equation solving. The new skill here is recognizing which second equation to manufacture.

Example

Suppose

$$f(x)+f(2-x)=10$$

and

$$f(2-x)-f(x)=2-2x.$$

Add the equations:

$$2f(2-x)=12-2x.$$

Then $f(2-x)=6-x$, and substituting back gives

$$f(x)=x+4.$$

5. Integer-domain propagation: when a functional equation creates a step rule

On the integers, a substitution may create a recurrence-like step. That does not make the original problem a sequence problem.

Example

For all integers m, n ,

$$f(m+n)=f(m)+f(n)+2mn$$

and $f(1)=1$.

Set $m=0$:

$$f(0)=0.$$

Set $n=1$:

$$f(m+1)=f(m)+1+2m.$$

This step relation propagates through the integers. It suggests $f(m)=m^2$, and the original equation verifies it:

$$(m+n)^2=m^2+n^2+2mn.$$

The step relation is a consequence of the functional equation. The proof finishes in the original two-variable relation.

6. Functional equation vs recurrence

These two can look similar.

A recurrence such as

$$a_{n+1}=a_n+2n+1$$

defines or constrains an indexed sequence.

A functional equation such as

$$f(m+n)=f(m)+f(n)+2mn$$

constrains one function at many related inputs.

If the domain is integers, setting $n=1$ can create a recurrence-like rule for $f(m)$. Use it, but remember where it came from and verify the final formula in the original equation.

7. A formula that fits values is only a conjecture

Suppose small calculations give

$$f(0)=1, f(1)=2, f(2)=3, f(3)=4.$$

The pattern $f(x)=x+1$ is plausible. It is not proved.

A proof must show the rule holds for every allowed input, usually by:

1. deriving it directly from strategic substitutions, or
2. substituting a candidate into the original functional equation and proving it satisfies every condition.

Finite data cannot certify an all-input statement.

8. When equal outputs force equal inputs

First use the plain-language test: if $f(a)=f(b)$, can the equation force $a=b$? When this is true for all allowed inputs, the function is called injective.

Consider, for all real x, y ,

$$f(x+f(y))=f(x)+y.$$

Suppose $f(a)=f(b)$. Then for any real x ,

$$x+f(a)=x+f(b),$$

so the left sides are equal. The equation gives

$$f(x)+a=f(x)+b,$$

hence $a=b$.

The equation itself proves the equal-output test; the word **injective** only names that result.

If you see xy , $mn+1$, $(x-1)y$, or another product-like argument:

First question: can one variable be 0 or 1 so the argument becomes a known constant?

Return partner

If you see $f(c-x)$ together with $f(x)$:

First line: write the same equation with x replaced by $c-x$.

Check that applying the replacement twice returns to x .

Two companion values

If the equations contain the same two unknown function values:

First line: name them temporarily, then add/subtract/eliminate.

Integer domain

If the equation contains $m+n$, $m-n$, or a shift:

First question: what do 0, 1, and -1 produce?

A unit-step relation may propagate all integer values.

Equal outputs

If you need to show equal outputs force equal inputs (the property later called injective):

First line: assume $f(a)=f(b)$ and use the given equation to see whether it forces $a=b$.

Every target output

If you need to show every target value occurs (the property later called surjective):

First line: take an arbitrary target t and try to construct an input whose output is t .

Three contrasts

- Random substitution asks "what happens at 2?" Strategic substitution asks "what removes the product/partner?"
- A recurrence generates indexed terms. A functional equation relates a function at transformed inputs.
- A table suggests a formula. The original equation proves it.

Stop checks

Before accepting a solution:

- all substitutions legal?
- any hidden division by zero?
- partner equation really equivalent?
- enough equations for the unknown function values?
- candidate checked on the full stated domain?

Recognition and First-Line Lab

For each item, write only the best first move or choose the best option. Do not complete a long solution unless needed to justify the choice.

Recognition

1. For integers, an equation contains $f(mn+1)$. Which is the strongest first test?
A. $m=7$ B. $m=0$ C. $m=n$ D. large positive m
2. For all real x , an equation contains $f(8-x)$ and $f(x)$. What structure matters?
A. periodicity B. the partner map returns to x after two uses C. divisibility D. ordering
3. You know values at x and $5-x$ occur in one equation. What should you manufacture?
A. a graph B. the equation at $5-x$ C. ten sample values D. a sequence table
4. A relation is stated only for integers. Which substitution needs no extra justification?
A. $x=1/2$ B. $x=\sqrt{2}$ C. $x=-1$ D. $x=\pi$
5. Five computed values fit $f(n)=n+1$. What is still missing?
A. arithmetic B. proof for every allowed input C. another table D. a graph
6. A problem asks whether equal outputs always force equal inputs. Which start is legitimate?
A. assume the function is increasing B. assume $f(a)=f(b)$ and try to force $a=b$ C. draw a sketch D. test three values
7. A problem asks whether every real number occurs as an output. Which start is legitimate?
A. pick arbitrary target t and construct an input B. test positive outputs C. assume continuity D. assume equal outputs force equal inputs

8. Which object is a recurrence rather than a functional equation?

A. $f(x+y)=f(x)+f(y)$ B. $a_{(n+1)}=2a_n+1$ C. $f(3-x)+f(x)=7$ D. $f(xf(y))=x+y$

First lines

9. For real x , $3f(4-x)+2f(x)=x+7$. Write the partner equation.

10. For integers m,n , $f(m+n)=f(m)+n$ and $f(0)=5$. Write the substitution that determines $f(n)$ immediately.

11. For integers, $f(m+n)=f(m)+f(n)+mn$. Write the substitution that reveals $f(0)$.

12. Suppose $f(x+f(y))=f(x)+y$ for all reals and you want to prove that equal function values force equal inputs. Write the assumption involving two inputs.

13. For all reals, $f(x)+f(2-x)=10$ and $f(2-x)-f(x)=2-2x$. Write the equation-combination move.

14. A proposed formula is $f(x)=x^2+1$ for $f(x+y)=f(x)+f(y)+2xy-1$. Write the verification line to check.

15. For integers, a derived rule is $f(n+1)=f(n)+2n+1$ and $f(0)=0$. State one thing you must still do before claiming the original two-variable functional equation is solved.

16. The equation contains $f(1-xy)$ on an integer domain. Name two legal structural substitutions worth testing first.

Practice and Transfer Bank

Try each item before looking at support. The stages reduce support gradually, but the method names are not supplied inside the questions.

Stage 0

1. For all real x , $f(x)+f(0)=x+2$. Find $f(7)$.

2. For all real x ,
 $f(x)+f(1-x)=5$ and
 $f(x)-f(1-x)=2x-1$.
Find $f(3)$.

3. For all real x , $2f(4-x)+f(x)=x+4$. Find $f(1)$.

4. For all integers m,n , $f(m+n)=f(m)+f(n)$ and $f(1)=3$. Find $f(-4)$.

Stage 1

5. A function $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(x+1)=f(x)+2$ and $f(0)=1$. Explain why these facts alone do not determine $f(x)$ for every real x .

6. For all positive real x,y , $f(xy)=x f(y)+y f(x)$. Determine $f(1)$.

7. For all integers m,n ,
 $f(m+n)=f(m)+f(n)+2mn$
and $f(1)=1$.
Find $f(6)$.

8. For all real x , $f(x)+f(2-x)=10$. Explain why the equation alone does not determine $f(5)$ uniquely.

Stage 2

9. For all real x ,
 $f(x)+f(2-x)=10$,
 $f(2-x)-f(x)=2-2x$.
Find $f(5)$.

10. For all real x , $3f(5-x)+2f(x)=x+10$. Find $f(-2)$.

11. For all real x , $4f(1-x)+f(x)=3x+1$. Find $f(7)$.

12. For all integers m,n ,
 $f(m+n)=f(m)+f(n)+mn$
and $f(1)=1$.
Find $f(8)$.

Stage 3

13. A function on the reals satisfies $f(x+f(y))=f(x)+y$ for all real x,y . Prove that equal function values force equal inputs. (This property is called injective.)

14. For the same equation, prove that every real target occurs as an output by constructing an input for an arbitrary target. (This property is called surjective.)

15. For all integers m, n , $f(m-n)=f(m)-n$ and $f(0)=3$. Find $f(-11)$.

16. For all real x , $2f(x)+f(3-x)=x+9$. Determine $f(20)$.

Stage 4

17. For all integers m, n ,

$$f(m+n)=f(m)+f(n)+mn$$

$$\text{and } f(0)=0, f(1)=1.$$

Derive a formula for $f(n)$ for every integer n , then verify it in the original equation.

18. A student computes $f(0)=1, f(1)=2, f(2)=3, f(3)=4$ and concludes $f(n)=n+1$ for all integers. Give a precise reason this conclusion is incomplete and state what would complete it.

19. For all real x , $3f(x)+f(4-x)=4x+4$. Determine $f(7)$ without guessing a formula from sample values.

20. A proposed function is $f(x)=x^2+1$. Verify directly that it satisfies

$$f(x+y)=f(x)+f(y)+2xy-1$$

for all real x, y .

Mixed Mastery Test

Work without method labels or hints. State the domain-sensitive first move clearly.

1. For all real x , $2f(x)+f(3-x)=x+9$. Find $f(10)$.

2. For all integers m, n ,

$$f(m+n)=f(m)+f(n)+mn$$

$$\text{and } f(1)=1.$$

Find $f(9)$.

3. For all real x , $f(x)+f(2-x)=6$. Is $f(5)$ uniquely determined? Justify.

4. Suppose $f(x+f(y))=f(x)+y$ for all real x, y . Prove that equal function values force equal inputs.

5. For all integers m, n , $f(m-n)=f(m)-n$ and $f(0)=4$. Find $f(13)$.

6. For all real x , $3f(x)+f(4-x)=4x+4$. Find $f(7)$.

7. A function on the integers satisfies $f(n+1)=f(n)+2n+1$ and $f(0)=0$. Explain why this is a recurrence-like consequence that may help with an integer functional equation, but is not by itself proof of an unrelated two-variable equation.

8. Verify whether $f(x)=x^2+1$ satisfies

$$f(x+y)=f(x)+f(y)+2xy-1$$

for every real x, y .

9. For all real x ,

$$f(x)+f(1-x)=5,$$

$$f(x)-f(1-x)=2x-1.$$

Find $f(-4)$.

10. A student tries $x=2, 3, 4$ in a functional equation and sees a neat pattern. State the test that separates a useful conjecture from a proof.